

Comparative Simulations of Novel Fault Recovery Strategies using Plug-in Electric Vehicles on the Grid Transient Stability

Bagaskara Primastya Putra^a and Adha Imam Cahyadi^b

Department of Electrical and Information Engineering, Faculty of Engineering, Universitas Gadjah Mada, Yogyakarta, Indonesia

ARTICLE HISTORY

Compiled May 26, 2023

ABSTRACT

Plug-in electric vehicle (PEV) usage has increased in the past decade to reduce fossil fuel consumption and greenhouse emissions. To exchange power between the electrical grid and the PEVs, vehicle-to-grid (V2G) and grid-to-vehicle (G2V) technologies have been developed. One of the challenges of incorporating PEVs into the grid is maintaining the grid stability. To solve this problem, V2G has been applied to deliver power from the PEV's batteries back to the grid. However, most research on V2G and G2V neglected the transient stability of the electrical grid and generators. In this paper, we analyzed the effect of three V2G/G2V configurations: a local frequency fault recovery control, a constant battery current V2G, and a constant battery current G2V on the transient stability of a modified IEEE-3 bus grid system when subject to short-circuit faults. An interesting finding was found that operating the constant battery current V2G on only one bus outperforms the local frequency control in all buses in terms of critical clearing time (CCT), thus providing a better grid transient stability. On the other hand, the G2V mode decreases the grid transient stability as indicated by a shorter CCT and a gradual drop of rotor speeds below the nominal speed.

KEYWORDS

electric, vehicle, grid, transient stability, fault

1. Introduction

Plug-in electric vehicle (PEV) usage has increased in the past decade to reduce fossil fuel consumption and greenhouse emissions. However, implementing the infrastructure for PEVs has its own challenges [1]. One of these challenges is maintaining the stability of the power system to accommodate PEV charging stations. Ideally, there should be a balanced supply and demand of electrical power between the generators and the consumers to maintain stability. However, since PEVs are only used during certain times of the day and mostly charged through the rest of the day, power can be delivered from the PEVs back to the grid when needed. This concept is called vehicle-to-grid (V2G) and can be applied vice versa with grid-to-vehicle (G2V) through the use of bidirectional AC/DC converters.

Research on V2G and G2V have been conducted in [2–4]. Arancibia et al. [2] pro-

CONTACT Bagaskara Primastya Putra. Email: bagaskara98@mail.ugm.ac.id

CONTACT Adha Imam Cahyadi. Email: adha.imam@ugm.ac.id

posed a method to reduce the time required to charge multiple PEVs simultaneously. This configuration consists of two main systems: a **bidirectional AC/DC converter and a battery charger**. The converter is connected with an LCL filter to reduce total harmonic distortion (THD) and controlled using a phase-locked loop (PLL) with a cascade proportional integral (PI) controller in the dq frame. On the other hand, the battery charger is implemented as a DC/DC converter which can be controlled using a constant current (CC) or a constant voltage (CV) PI controller. This configuration has been simulated on MATLAB/Simulink with two PEVs connected to the charging station on different load configurations. The simulation results show proper dynamic behavior of the DC bus voltage, battery voltage, and battery current. They also found that smooth transition to V2G and reactive power compensation are possible.

Evode [3] proposed a similar V2G/G2V model to [2]. However, the simulations in [3] were conducted for different combinations of five PEVs connected to the charging station instead of only two PEVs in [2]. Not only are the charging station's and batteries' voltage, current, and power analyzed, Evode also analyzed the grid voltage, current, and power. **The simulations show that G2V, V2G, and possible auxiliary modes such as vehicle-to-vehicle (V2V), vehicle-to-load (V2L), and grid-to-load (G2L) are feasible.**

Olagbemi et al. [4] conducted a transient analysis of a V2G system under varying grid and vehicle load conditions. They wanted to investigate the effect of operational mode changes on the transients of the electrical grid, especially the effect of harmonic distortion. **The simulation results show that the LCL filter components should be carefully chosen to minimize harmonic distortion at both the DC and AC ports of the bidirectional converter.**

However, the research conducted in [2–4] mostly focused on the performance of the PEV charging station variables. Most of these research neglected the transient stability of the electrical grid and generators. Furthermore, the generators are only modelled as a three-phase voltage source without considering the dynamics of the swing equation. To solve this problem, Gajduk et al. [5] included the swing equation of the generators into the model. A local frequency-based controller has been proposed to control the output power of the charging station. This controller has been simulated on several IEEE bus cases and the results show that the V2G/G2V system improves both the steady state and transient stability of the power system. **A better steady state stability was shown with a decrease of eigenvalues.** On the other hand, **a better transient stability was shown with voltage and frequency oscillations that dampen out faster, longer critical clearing time (CCT) after a fault, and a larger region of asymptotic stability (RAS).** However, the charging stations are only modelled as injected power without modelling the components such as the bidirectional converter and battery chargers.

In this paper, we propose the transient stability analysis of V2G and G2V operations on a modified IEEE-3 case electrical grid. It includes the modelling of the generators' dynamics using swing equations as well as the bidirectional AC/DC converter and battery chargers. Furthermore, three-phase short circuit faults on the electrical grid will be simulated in MATLAB/Simulink with the following configurations:

- (1) the electrical grid itself without any V2G nor G2V,
- (2) local frequency fault recovery control from [5],
- (3) our proposed constant battery current V2G, and
- (4) a constant battery current G2V.

We would also like to find out the effects of the active bus locations on the transient stability. This will be accomplished by simulating different combinations of V2G/G2V bus locations and observing their transient response and CCT.

2. Power System Model

2.1. Electrical Grid

In this paper, the electrical grid is modelled after the IEEE-3 bus case [6] which is a three-generator three-bus system. Originally the grid in [6] operates on a base voltage of 345 kV, a base apparent power $S_{base} = 100$ MVA, and a nominal grid frequency $f_{grid} = 60$ Hz. However, for the purpose of our simulations, the base apparent power is modified to $S_{base} = 1$ MVA while the nominal grid frequency is modified to $f_{grid} = 50$ Hz. The reason for this modification is that the V2G/G2V design in [2] operates in a lower power and frequency compared to the IEEE-3 grid [6]. If the grid's parameters are still the same as in [6], then there would need to be more batteries and controllers installed in the V2G/G2V system which can lengthen the time required to run the simulations. Therefore, in order for the V2G/G2V system to have a more significant effect on the grid, the grid base apparent power and frequency is lowered to the level of the V2G/G2V system.

Each bus has one RL load connected to it with the per unit (p.u.) power demand at the i^{th} bus S_i defined as $S_1 = 1.5 + j.45$, $S_2 = 1.0 + j.3$, and $S_3 = 12.4 + j2.5$. Moreover, the transmission line impedance between the i^{th} and j^{th} bus $Z_{i,j}$ in p.u. values are defined as $Z_{1,2} = j.46$, $Z_{1,3} = j.26$, and $Z_{2,3} = j.806$. Compared to the original grid in [6], our grid configuration has a V2G/G2V system attached to each bus as shown in Figure 1. To accomodate the voltage difference between the grid and the V2G/G2V system, a 345/0.8 kV step-down transformer is attached to each bus labelled by T_i .

A three-phase short circuit fault located near bus 1 at the transmission line between bus 1 and 2 will then be simulated at $t = 0.5$ seconds over various fault clearing times (FCTs) to obtain the critical clearing time (CCT). These time parameters are used to determine the transient stability of the system when subject to faults. The FCT is the time when the fault is cleared. If the FCT is less than or equal to the CCT, the system remains stable. However, if the FCT is greater than the CCT, the system will become unstable [7].

This approach of transient stability analysis has also been implemented in [8] and [9]. A general overview of grid transient stability analysis which includes the time-domain method and direct method has been proposed in [8]. One of the proposed time-domain method is by gradually increasing the FCT until the rotor angle and rotor speeds lose synchronism. Similarly in [9], the transient stability of a two-generator six-bus grid has also been analyzed using the Lyapunov function and then verified using simulation in MATLAB/Simulink.

2.2. Generators

The i^{th} generator is modelled electrically as a three-phase internal voltage source E_i connected in series to a three-phase impedance Z_i . On the mechanical side, the generator is modelled using swing equations defined as

$$\dot{\delta}_i = \omega_i \quad (1)$$

$$\dot{\omega}_i = \frac{1}{M_i}(-D_i\omega_i + P_{m_i} - P_{e_i}) \quad (2)$$

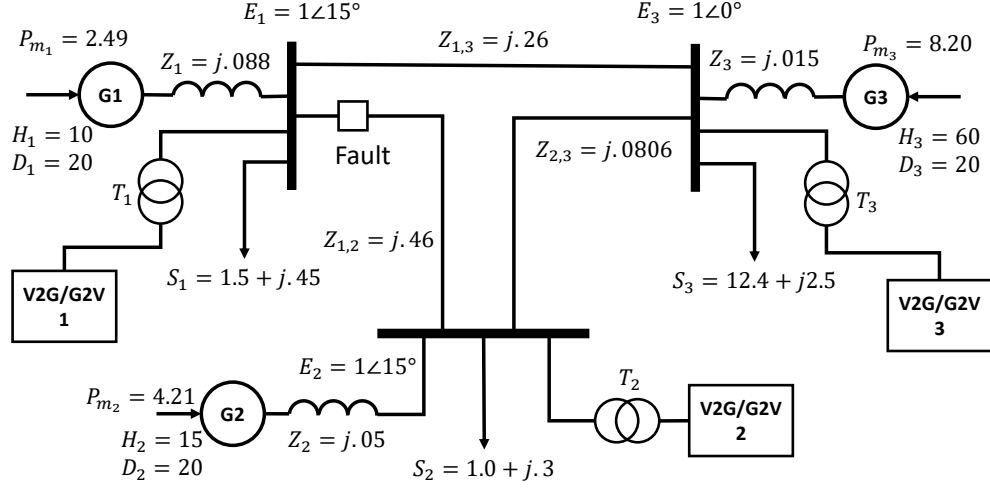


Figure 1. IEEE-3 bus case system with V2G/G2V installed at each bus.

where δ_i is the rotor angle, ω_i is the rotor angular speed, D_i is the damping coefficient, M_i is the rotor inertia constant, P_{m_i} is the mechanical power input, and P_{e_i} is the electrical power output.

For the purpose of our transient stability analysis, the generators are run in an open-loop configuration without any governor nor automatic voltage regulator (AVR). However, the magnitude of the generator's input power P_{m_i} and internal voltage source E_i are maintained constant. This configuration is chosen such that we can analyze the V2G/G2V's effect on the grid stability better than if the generators are run in a closed-loop configuration.

3. V2G/G2V System

The V2G/G2V system consists of the following subsystems: an LCL filter, a bidirectional AC/DC converter, and the battery chargers as shown in Figure 2. Each battery charger can be connected to one PEV battery. Instead of only two battery chargers based on [2], five battery chargers will be installed in the V2G/G2V system. This number of battery chargers is chosen because the power level in the IEEE-3 grid is in the orders of MVA. Furthermore, the configurations and parameters are based on [2], unless some modifications are further stated in sections 3.1-3.3.

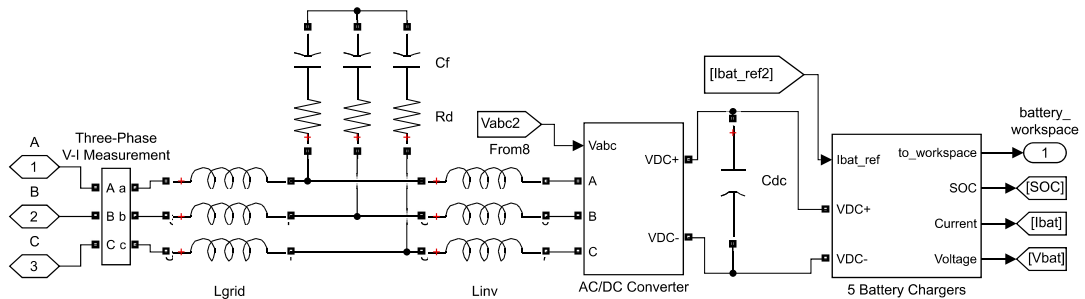


Figure 2. Proposed V2G/G2V system.

3.1. LCL Filter

The LCL filter is installed between the transformer and the bidirectional AC/DC converter as shown in Figures 1 and 2. This filter is intended to reduce harmonics caused by power electronics switching. In this paper, five batteries will be installed in the V2G/G2V system. However, the original V2G/G2V design in [2] only supports two batteries. Therefore, the filter components need to be reconfigured from the original values: $L_{inv} = 0.48$ mH, $L_{grid} = 0.69$ mH, and $C_f = 165$ μ F to the modified values: $L_{inv} = 48$ μ H, $L_{grid} = 69$ μ H, and $C_f = 1.65$ mF.

3.2. Bidirectional AC/DC Converter

The bidirectional AC/DC converter consists of six insulated-gate bipolar transistors (IGBTs) as shown in Figure 3. These IGBTs are controlled using pulse width modulation (PWM) signals from a cascade proportional integral (PI) control in the $dq0$ -frame. This controller consists of an outer voltage loop and inner current loop with some feedforward terms. The $dq0$ voltages and currents are obtained from their three phase abc counterpart using Park's transformation [10]. To synchronize the voltage between the converter and the grid, the phase-locked loop (PLL) is utilized. As mentioned in section 3.1, the original converter parameters in [2] were only rated for two battery chargers which need to be reconfigured for five battery chargers as proposed in this paper. Hence, the DC-link capacitor is increased from $C_{dc} = 18$ mF to $C_{dc} = 1.8$ F.

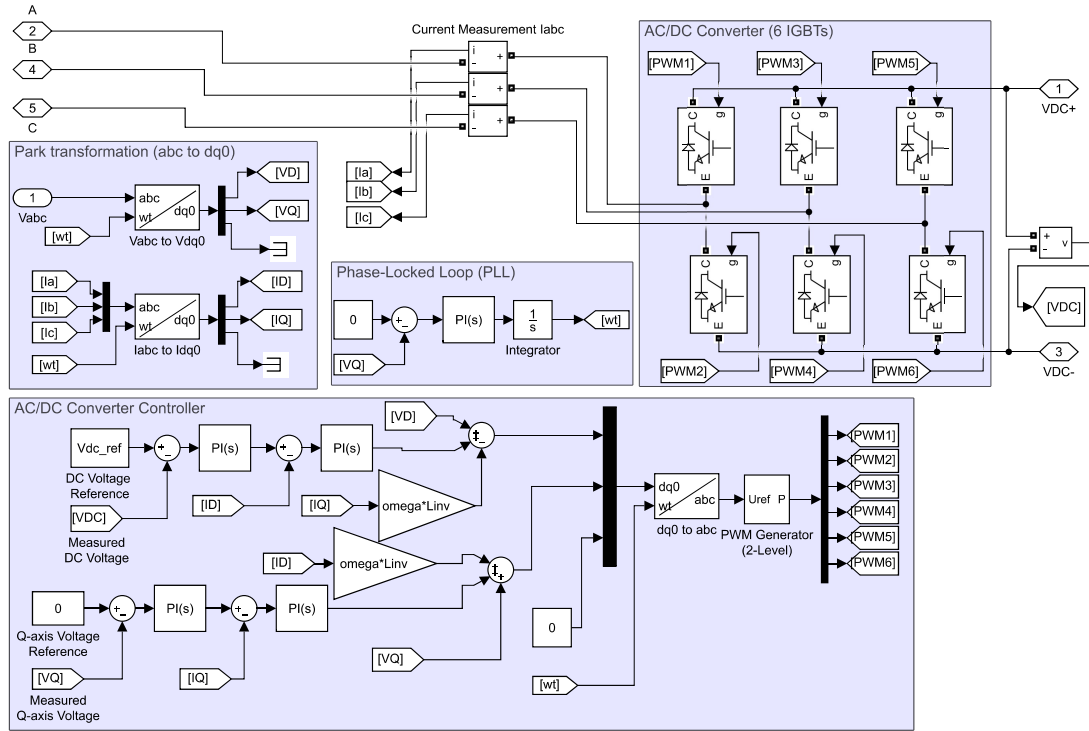


Figure 3. AC/DC converter circuit and controller.

3.3. Battery Charger

The battery charger consists of a buck-boost converter circuit. This circuit is comprised of two IGBTs, each of them to control the buck mode and the boost mode in a complimentary manner as shown in Figure 4.

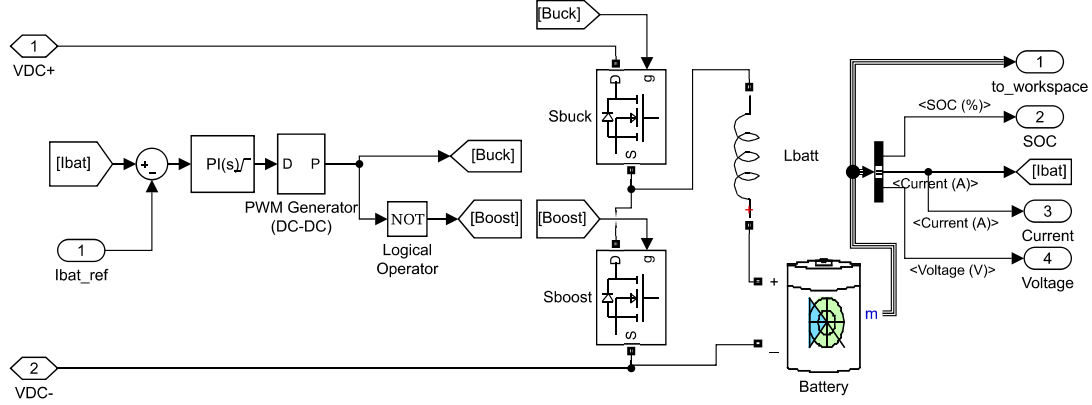


Figure 4. Battery charger circuit and controller.

Some modifications were made to the original battery charger design in [2]. First, the battery current sensor is installed such that a measured battery current i_{bat} is positive when the current flows out of the battery's positive terminal. Based on this configuration, a positive i_{bat} refers to the V2G mode, while a negative i_{bat} refers to the G2V mode.

Then instead of using the PI controller design from [2] which includes feedforward terms from the DC link and battery voltage, a simple battery current PI controller is utilized. This configuration is chosen because when we tried simulating the original configuration from [2], it results in the battery current not converging to its reference value, most likely due to the feedforward terms. Furthermore, five of these battery chargers are connected in parallel and given the same battery reference current signal as shown in Figure 5.

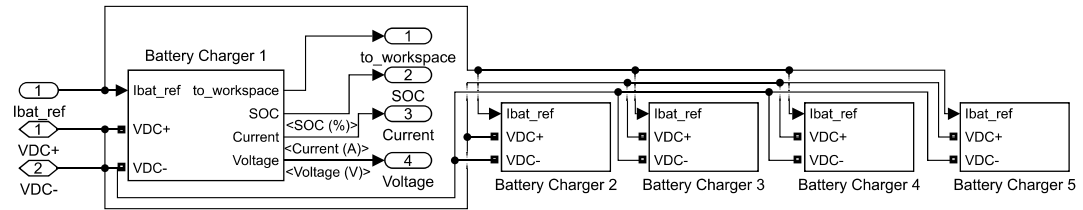


Figure 5. Battery charger circuit and controller.

3.4. Local Frequency Fault Recovery Control

In [5], a local frequency control strategy has been proposed to increase the grid transient stability by transferring power to and from the grid proportional to the frequency

deviation measured at the local bus. This control strategy can be formulated as

$$P_i^{PEVG} = \begin{cases} -0.1 \cdot h_{PEV}, & \text{if } \Delta f_i \leq -0.1 \text{ Hz} \\ \Delta f_i \cdot h_{PEV}, & \text{if } -0.1 \text{ Hz} < \Delta f_i \leq 0.1 \text{ Hz} \\ 0.1 \cdot h_{PEV}, & \text{if } \Delta f_i > 0.1 \text{ Hz} \end{cases} \quad (3)$$

$$\Delta f_i = f_i - f_{grid} \quad (4)$$

where P_i^{PEVG} is the aggregate power of all PEVs at the i^{th} bus, h_{PEV} is a control parameter, Δf_i is the electric frequency deviation at the i^{th} bus, f_i is the measured frequency at the i^{th} bus, and f_{grid} is the base grid frequency. This control strategy not only increased the CCT, but also reduced the post-fault voltage and frequency fluctuations. Unfortunately, the simulations conducted in [5] only considered the power flow of the V2G/G2V without modelling the dynamics of the converters and battery chargers. It was unclear how it would be implemented if the dynamics of the converters and battery chargers were also modelled.

In order for the control strategy in (3) to be implemented in our work, the control strategy in (3) can be modified to

$$I_{bat}^{ref} = \begin{cases} -0.1 \cdot h_{PEV}, & \text{if } \Delta f_i \leq -0.1 \text{ Hz} \\ \Delta f_i \cdot h_{PEV}, & \text{if } -0.1 \text{ Hz} < \Delta f_i \leq 0.1 \text{ Hz} \\ 0.1 \cdot h_{PEV}, & \text{if } \Delta f_i > 0.1 \text{ Hz.} \end{cases} \quad (5)$$

This modified control strategy controls the battery reference current I_{bat}^{ref} proportional to the frequency deviation Δf_i but saturated at ± 0.1 Hz.

3.5. Constant Battery Current V2G

In this paper, we propose a simpler fault recovery control strategy consisting of a constant battery reference current V2G that is triggered by a moving root mean square (RMS) bus voltage fault detection. The moving RMS of the bus voltage is given by

$$V_{RMS}[k] = \sqrt{\frac{1}{N} \sum_{i=k-N+1}^k V^2[i]} \quad (6)$$

where $V[i]$ is the i^{th} sample of the measured bus voltage, $N = T_{grid}/T_s$, $T_{grid} = 1/f_{grid}$, and T_s is the sampling period of the simulation. The simulation sampling period is set to $T_s = 10^{-5}$ seconds which still satisfies the Nyquist frequency given the highest frequency from the PWM switching frequency is 5000 Hz.

Moreover, since there are three phases in the electrical grid, the average between each phase's moving RMS will be utilized to detect a fault, that is

$$V_{RMS,abc}[k] = \frac{(V_{RMS,a}[k] + V_{RMS,b}[k] + V_{RMS,c}[k])}{3}. \quad (7)$$

If this average value is below a certain threshold $V_{faultThresh}$, then the battery reference

current of the battery charger will be set to a constant positive current given by

$$I_{bat}^{ref} = \begin{cases} I_{batFault}^{ref}, & \text{if } V_{RMS,abc} < V_{faultThresh} \\ 0, & \text{otherwise.} \end{cases} \quad (8)$$

This non-zero battery reference current will be held constant until the rotor speed oscillations dampen out. Here, $I_{batFault}^{ref}$ is set to 200 A which is the original maximum current rating from [2]. On the other hand, $V_{faultThresh}$ is tuned to 455 V based on the observations of previous fault simulations. Furthermore, the complete MATLAB/Simulink simulations can be accessed in Github ¹.

4. Simulation Results

The design and configurations elaborated in sections 2 and 3 will then be simulated in MATLAB/Simulink. More specifically, these configurations include:

- (1) the electrical grid itself without any V2G nor G2V,
- (2) the local frequency fault recovery control derived in section 3.4,
- (3) our proposed constant battery current V2G derived in section 3.5, and
- (4) a constant battery current G2V.

These simulations will then be conducted on different combinations of V2G/G2V bus locations: bus 1, bus 2, bus 3, and all 3 buses together. For example, the bus 1 V2G/G2V configuration means that only bus 1 has the V2G/G2V system installed while buses 2 and 3 do not.

For a fair comparison, the battery current is limited to $[-200, 200]$ A. More specifically, the local frequency fault recovery control has the h_{PEV} value set to -2000 , the proposed constant battery current V2G has the $I_{batFault}^{ref}$ value set to 200 A, while the G2V has the I_{bat}^{ref} value set to -200 A.

4.1. Without V2G/G2V

In this configuration, the electrical grid does not have any V2G/G2V system connected to it. It can be observed in Figure 6a that the rotor angles increase synchronously and the rotor speed oscillations in Figure 6b decay and converge to the nominal speed of 1 p.u. These results indicate that this system is stable given a FCT of 0.241 seconds. However, when the FCT is set to 0.242 seconds, the rotor angles and rotor speeds deviate from each other as shown in Figures 7a and 7b, respectively. These deviations indicate that this system is unstable given a FCT of 0.242 seconds. Therefore, the CCT for the configuration where V2G/G2V is not connected to the electrical grid is 0.241 seconds.

As we can see in Figures 6a and 7a, the rotor angles both in the stable and unstable configuration resemble a ramp function. However, the only difference between these configurations is that the unstable rotor angles shown in Figure 7a deviate from each other as opposed to the stable rotor angles shown in Figure 6a that increase synchronously. On the other hand, the rotor speeds in Figures 6b and 7b show a better insight on the transient stability indicated by the oscillations and convergence of the rotor speeds. Therefore, from this point on we will only show the rotor speed results.

¹github.com/BagaskaraPutra/v2g-g2v-ieee3-transient

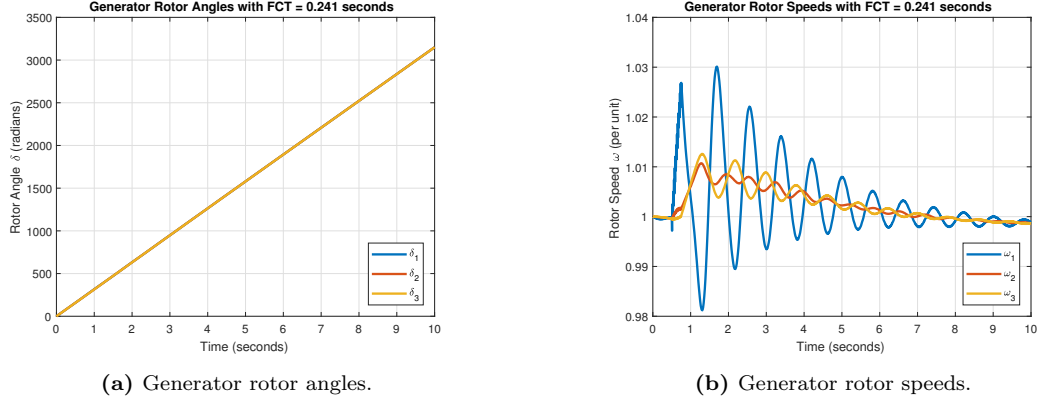


Figure 6. Generator rotor angles and speeds without V2G/G2V with FCT= 0.241 seconds. (Stable)

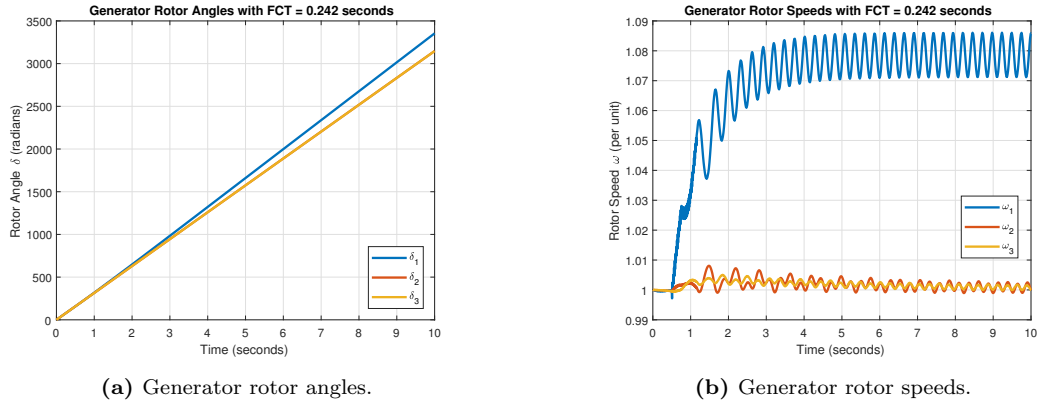
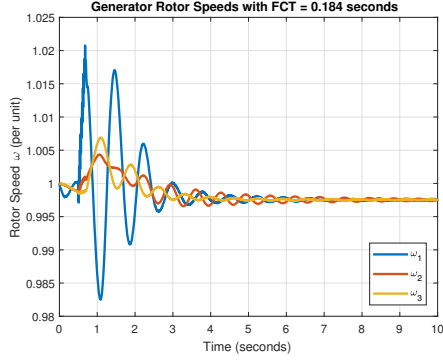


Figure 7. Generator rotor angles and speeds without V2G/G2V with FCT= 0.242 seconds. (Unstable)

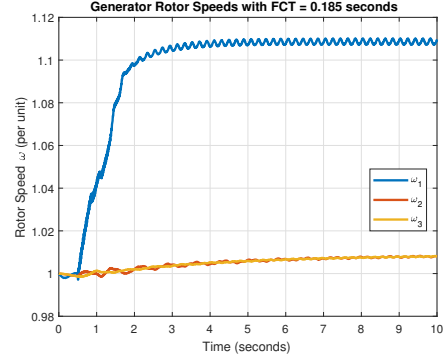
4.2. Local Frequency Fault Recovery Control

In this configuration, the V2G/G2V system is connected to the electrical grid and operated using the local frequency control strategy previously derived in section 3.4. Compared to the configuration without V2G/G2V, the CCT when only bus 1 has its V2G/G2V system active actually decreases to 0.184 seconds as shown in Figures 8a and 8b. However, when the V2G/G2V system is active in either bus 2 or 3, the CCT improves to 0.242 seconds as shown in Figures 8c-8f. Moreover, the best active bus configuration for the local frequency control is all of the buses which results in a CCT of 0.243 seconds as shown in Figures 8g and 8h.

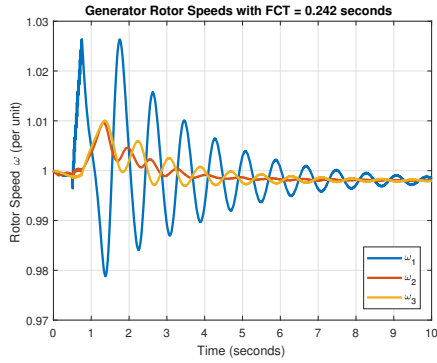
A comparison of the resulting CCTs is summarized in Table 1. It can be observed that applying the local frequency control at bus 1, where the fault is located, actually results in a worse CCT compared to the configuration without any V2G/G2V. On the other hand, applying the local frequency control on either bus 2 or 3, where the fault is not located, results in a better CCT. Moreover, applying the local frequency control on all buses results in a CCT improvement of 0.002 seconds or 0.83% compared to without any V2G/G2V.



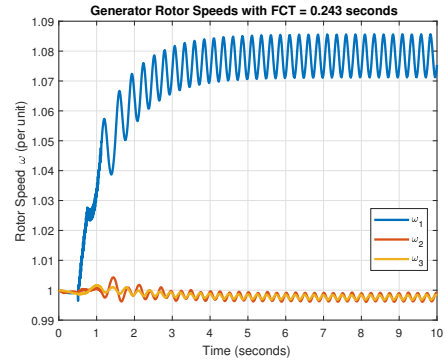
(a) Active bus= 1, FCT= 0.184 s (Stable)



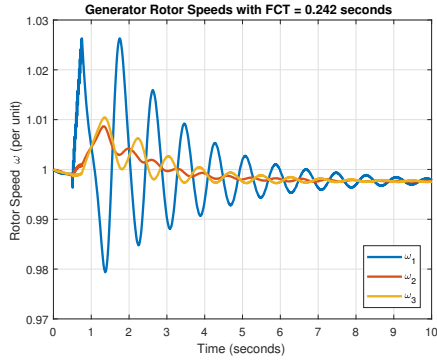
(b) Active bus= 1, FCT= 0.185 s (Unstable)



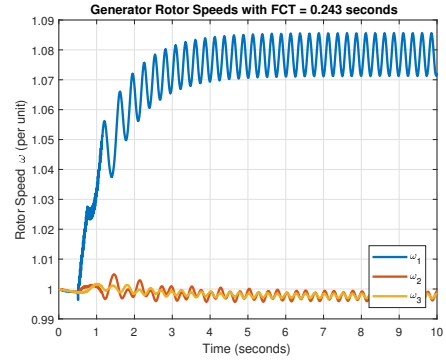
(c) Active bus= 2, FCT= 0.242 s (Stable)



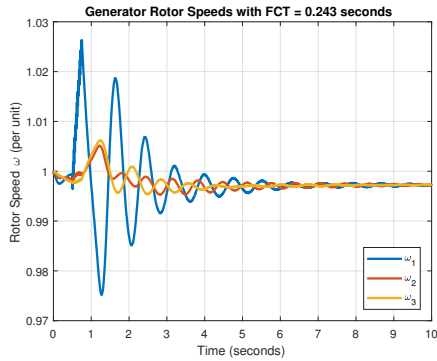
(d) Active bus= 2, FCT= 0.243 s (Unstable)



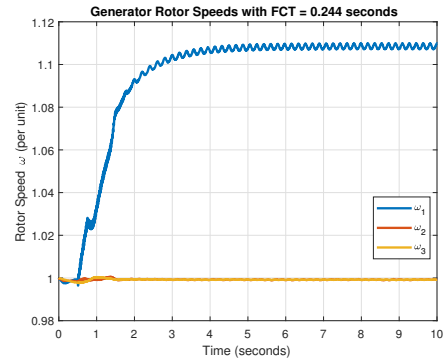
(e) Active bus= 3, FCT= 0.242 s (Stable)



(f) Active bus= 3, FCT= 0.243 s (Unstable)



(g) Active bus= 1, 2, 3, FCT= 0.243 s (Stable)



(h) Active bus= 1, 2, 3, FCT= 0.244 s (Unstable)

Figure 8. Effect of different FCTs and active bus locations of the local frequency fault recovery control on the generator rotor speeds.

Table 1. The local frequency control active bus effect on the critical clearing time (CCT)

No.	V2G/G2V configuration	Active V2G/G2V bus			CCT (seconds)	CCT improvement compared to without V2G/G2V	
		1	2	3		(seconds)	(%)
1	Without V2G/G2V				0.241	0	0
2	Local frequency control	✓			0.184	-0.057	-23.65
3	Local frequency control		✓		0.242	0.001	0.41
4	Local frequency control			✓	0.242	0.001	0.41
5	Local frequency control	✓	✓	✓	0.243	0.002	0.83

4.3. Constant Battery Current V2G

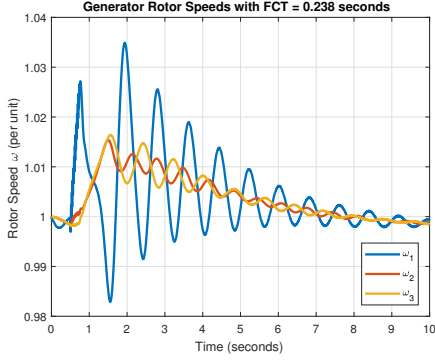
In this configuration, the V2G/G2V system is connected to the electrical grid and operated in the constant battery current V2G control strategy derived in section 3.5. Compared to the configuration without V2G/G2V, the CCT actually decreases to 0.238 seconds when the V2G/G2V at bus 1 is active as shown in Figures 9a and 9b. Interestingly, when either bus 2 or 3 are active, the CCT increases to 0.250 seconds as shown in Figures 9c - 9f. This result is by far the largest CCT out of all of the V2G/G2V configurations. Unfortunately, when all of the buses are active, the CCT is actually the same as the configuration without V2G/G2V. These results are also summarized in Table 2.

Comparing the CCT results of the constant battery current V2G in Table 2 with the local frequency control in Table 1, the CCT of the constant battery current V2G is generally greater than the local frequency control's CCT except for the configuration where all of the buses are active. However, upon further inspection on the rotor speeds, the local frequency control shown in Figure 8 dampens the oscillations better than the constant battery current V2G shown in Figure 9.

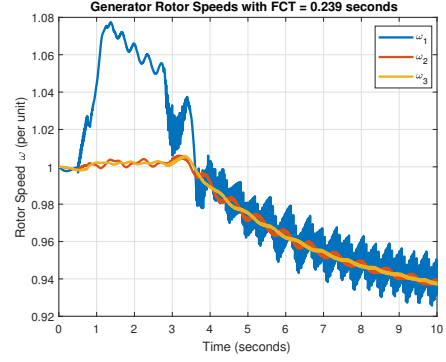
Nevertheless, the constant battery current V2G has the advantage of requiring fewer active buses and a simpler control strategy compared to the local frequency control. An example of this effect can be seen in Appendix A. In order to achieve a CCT of 0.242 seconds, the local frequency control in Figure A1 requires highly oscillating battery reference currents. These oscillation are due to the battery reference current being linear to the frequency deviation and may pose the risk of component damage in the battery chargers. Interestingly, to achieve a better CCT of 0.250 seconds, the constant battery current V2G shown in Figure A2 only requires a simpler battery reference current which resembles a step function.

Table 2. The constant battery current V2G active bus effect on the critical clearing time (CCT)

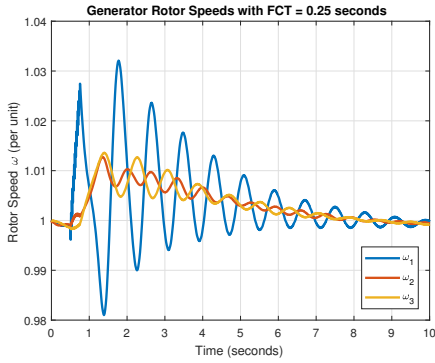
No.	V2G/G2V configuration	Active V2G/G2V bus			CCT (seconds)	CCT improvement compared to without V2G/G2V	
		1	2	3		(seconds)	(%)
1	Without V2G/G2V				0.241	0	0
2	Constant battery current V2G	✓			0.238	-0.003	-1.24
3	Constant battery current V2G		✓		0.250	0.009	3.73
4	Constant battery current V2G			✓	0.250	0.009	3.73
5	Constant battery current V2G	✓	✓	✓	0.241	0	0



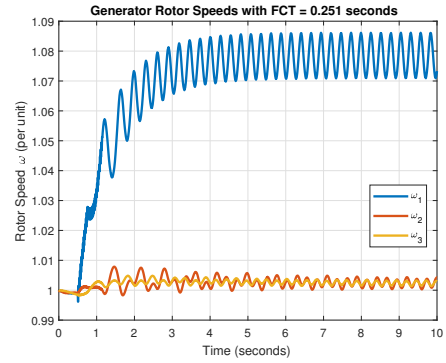
(a) Active bus= 1, FCT= 0.238 s (Stable)



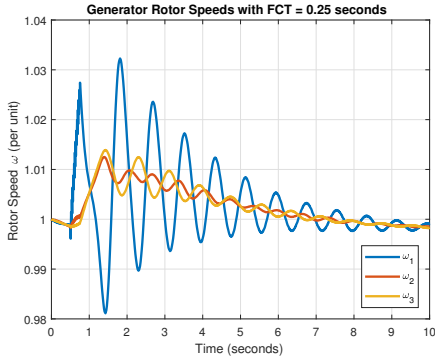
(b) Active bus= 1, FCT= 0.239 s (Unstable)



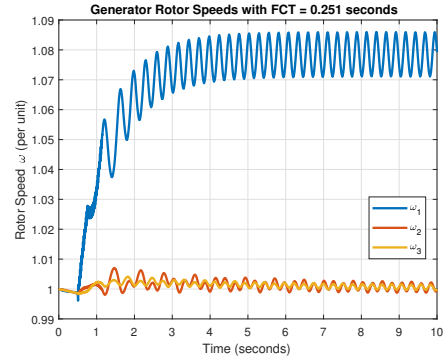
(c) Active bus= 2, FCT= 0.250 s (Stable)



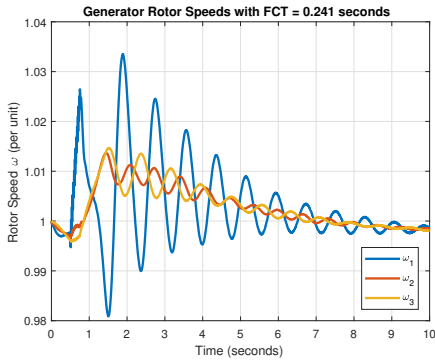
(d) Active bus= 2, FCT= 0.251 s (Unstable)



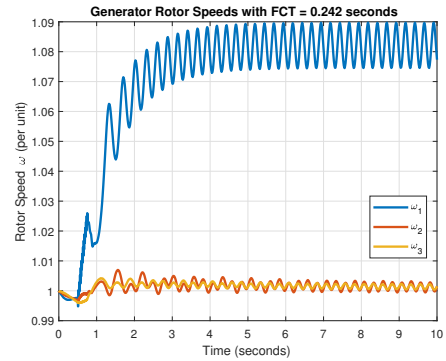
(e) Active bus= 3, FCT= 0.250 s (Stable)



(f) Active bus= 3, FCT= 0.251 s (Unstable)



(g) Active bus= 1, 2, 3, FCT= 0.241 s (Stable)



(h) Active bus= 1, 2, 3, FCT= 0.242 s (Unstable)

Figure 9. Effect of different FCTs and active bus locations of the constant battery current V2G on the generator rotor speeds.

4.4. G2V Connected to the Grid

In this configuration, the battery chargers are given a constant reference current of $I_{bat}^{ref} = -200$ A. This causes the V2G/G2V system to operate in the G2V mode, charging the batteries by withdrawing power from the grid. Compared to the configuration without V2G/G2V, it can be observed in Figures 10a and 10b that operating the G2V on bus 1 drastically decreases the CCT to 0.082 seconds. Interestingly, when the G2V is operated on either bus 2 or 3, the CCT only decreases to 0.238 seconds as shown in Figures 10c - 10f. Operating the G2V on either bus 2 or 3 only results in a 1.24% decrease in CCT compared to a 65.98% decrease in CCT when the G2V is operated on bus 1. Unfortunately, operating the G2V on all buses results in a CCT of 0.083 seconds or a 65.56% decrease in CCT as shown in Figures 10g and 10h. These CCT results are also summarized in Table 3.

It can also be observed that at the CCT, the rotor speeds in the G2V mode gradually drop below the nominal speed as shown in Figure 10. This result is different compared to the results of the configuration without V2G/G2V, the local frequency control, and the constant battery current V2G in sections 4.1-4.3 where the rotor speeds converge to the nominal speed. In the G2V mode, the increase in electrical power consumption exceeds the mechanical power input which results in a negative angular acceleration as shown in (2), thus gradually decreasing the rotor speeds.

Table 3. The G2V active bus effect on the critical clearing time (CCT)

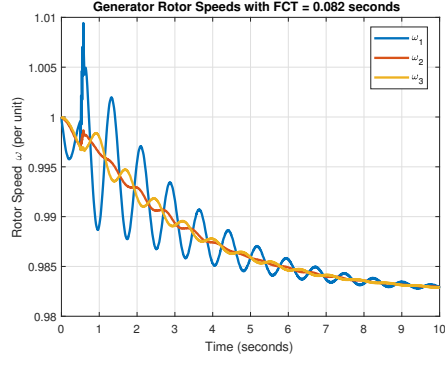
No.	V2G/G2V configuration	Active V2G/G2V bus			CCT (seconds)	CCT improvement compared to without V2G/G2V	
		1	2	3		(seconds)	(%)
1	Without V2G/G2V				0.241	0	0
2	G2V	✓			0.082	-0.159	-65.98
3	G2V		✓		0.238	-0.003	-1.24
4	G2V			✓	0.238	-0.003	-1.24
5	G2V	✓	✓	✓	0.083	-0.158	-65.56

4.5. Summary of CCT Results

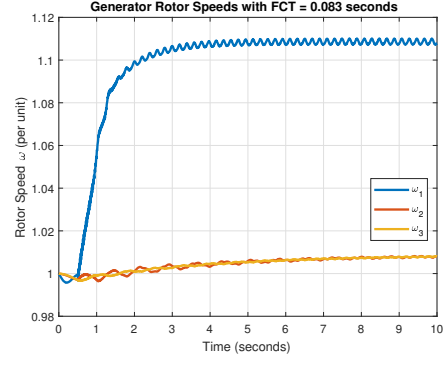
Finally, the CCT results elaborated in sections 4.1-4.4 can be summarized in a bar plot shown in Figure 11. It can be observed that the constant battery current V2G results in the best CCT (0.250 seconds) compared to other V2G/G2V configurations. More specifically, applying the constant battery current V2G on either bus 2 or 3 results in the best transient stability out of all the other V2G/G2V configurations.

However, operating the V2G/G2V in the G2V mode generally results in a decrease in CCT compared to the configuration without V2G/G2V. This means that the G2V mode generally results in a worse transient stability.

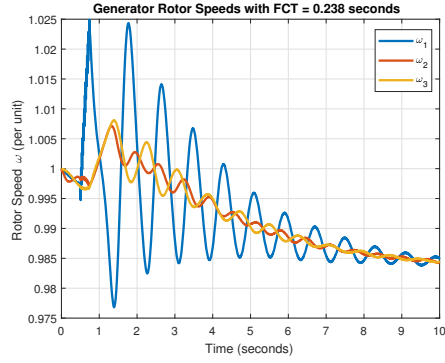
Another interesting finding is that applying any control strategy on bus 1 results in the worst CCT compared to other active buses of the same configuration. This result suggests that it is not advisable to operate the V2G/G2V at the location of the fault as it can result to a decrease in CCT and transient stability.



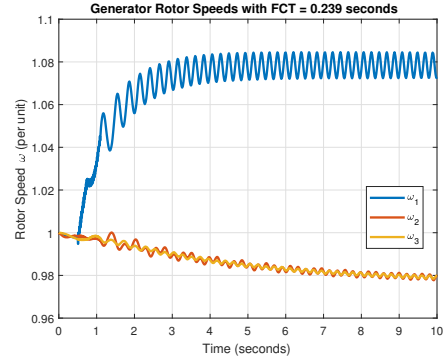
(a) Active bus= 1, FCT= 0.082 s (Stable)



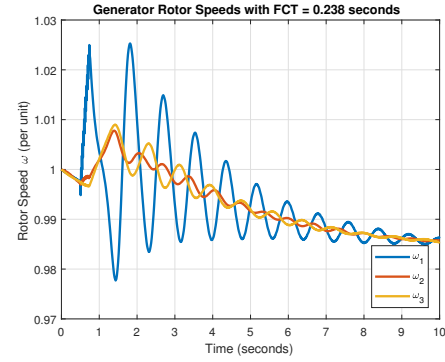
(b) Active bus= 1, FCT= 0.083 s (Unstable)



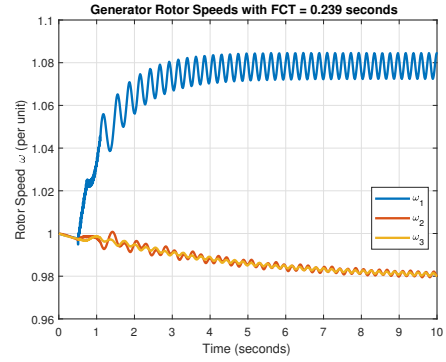
(c) Active bus= 2, FCT= 0.238 s (Stable)



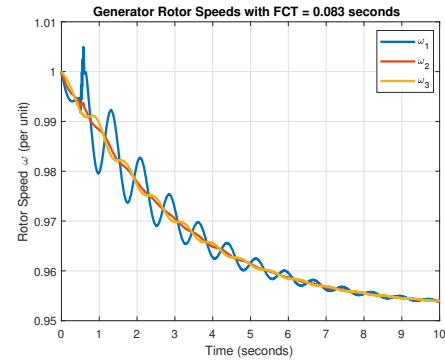
(d) Active bus= 2, FCT= 0.239 s (Unstable)



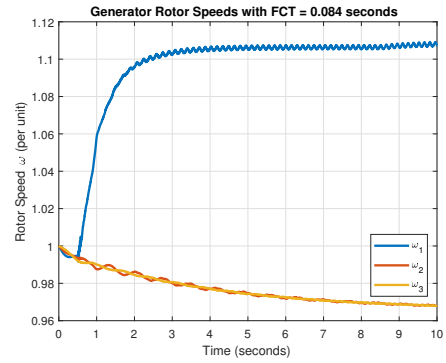
(e) Active bus= 3, FCT= 0.238 s (Stable)



(f) Active bus= 3, FCT= 0.239 s (Unstable)



(g) Active bus= 1, 2, 3, FCT= 0.083 s (Stable)



(h) Active bus= 1, 2, 3, FCT= 0.084 s (Unstable)

Figure 10. Effect of different FCTs and active bus locations of the G2V on the generator rotor speeds.

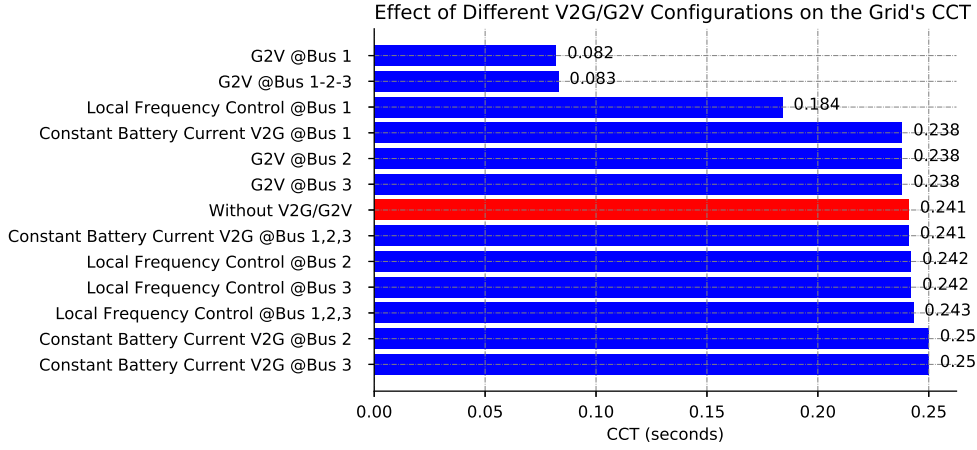


Figure 11. Effect of different V2G/G2V configurations on the electrical grid's CCT.

5. Conclusion

In this paper, numerical simulations have been conducted to analyze the effect of different PEVs V2G and G2V configurations on the transient stability of the modified IEEE-3 bus case when subject to short-circuit faults. Interestingly, operating the constant battery current V2G configuration at either bus 2 or 3 results in a larger CCT than operating the local frequency control on all buses. Therefore, to improve the grid transient stability during faults, it is recommended to operate the V2G/G2V in the constant battery current V2G configuration at the bus where the fault is not located.

On the contrary, the G2V configuration decreases the grid transient stability through a decrease of CCT and a gradual drop of generator rotor speeds below the nominal speed. Therefore, it is not recommended to charge the PEVs during the occurrence of faults in the electrical grid.

For future works, it is recommended to mathematically prove the fault recovery control strategies to obtain a better general insight on how these strategies can actually improve the electrical grid's transient stability. Variations of the fault locations, active bus locations, power levels, and control strategies are also considered for future works.

Disclosure statement

No potential competing interest was reported by the authors.

References

- [1] Veza I, Abas MA, Djamari DW, et al. Electric Vehicles in Malaysia and Indonesia: Opportunities and Challenges. *Energies*. 2022;15(7):1–24.
- [2] Arancibia A, Strunz K. Modeling of an electric vehicle charging station for fast DC charging. 2012 IEEE International Electric Vehicle Conference, IEVC 2012. 2012;(3).
- [3] Evode R. Modeling of electric grid behaviors having electric vehicle charging stations with G2V and V2G possibilities. *International Conference on Electrical, Computer, Communications and Mechatronics Engineering, ICECCME 2021*. 2021;(October):7–8.

- [4] Olagbemi OO, Gomez P, Meyer RT. Transient analysis of vehicle-to-grid systems incorporating plug-in electric vehicles. 2021 56th International Universities Power Engineering Conference: Powering Net Zero Emissions, UPEC 2021 - Proceedings. 2021;.
- [5] Gajduk A, Zdraveski V, Basnarkov L, et al. A Strategy for Power System Stability Improvement via Controlled Charge/Discharge of Plug-in Electric Vehicles. arXiv. 2015; (June). Available from: <http://arxiv.org/abs/1506.07097>.
- [6] Llamas A, La Ree Lopez JD, Mili L, et al. Clarifications of the BCU method for transient stability analysis. IEEE Transactions on Power Systems. 1995;10(1):210–219.
- [7] Sauer PW, Pai MA. Power System Dynamics and Stability. Prentice Hall; 1998.
- [8] Gajduk A, Todorovski M, Kocarev L. Stability of power grids: An overview. European Physical Journal: Special Topics. 2014;223(12):2387–2409.
- [9] Pratama GNP, Cahyadi AI. Lyapunov analysis of two imbalanced power system areas. ECTI Transactions on Electrical Engineering, Electronics, and Communications. 2021; 19(3):312–321.
- [10] MathWorks. Implement abc to dq0 transform - Simulink ; 2017. Available from: <https://www.mathworks.com/help//sps/ref/parktransform.html>.

Appendix A. Battery Reference Current Comparison

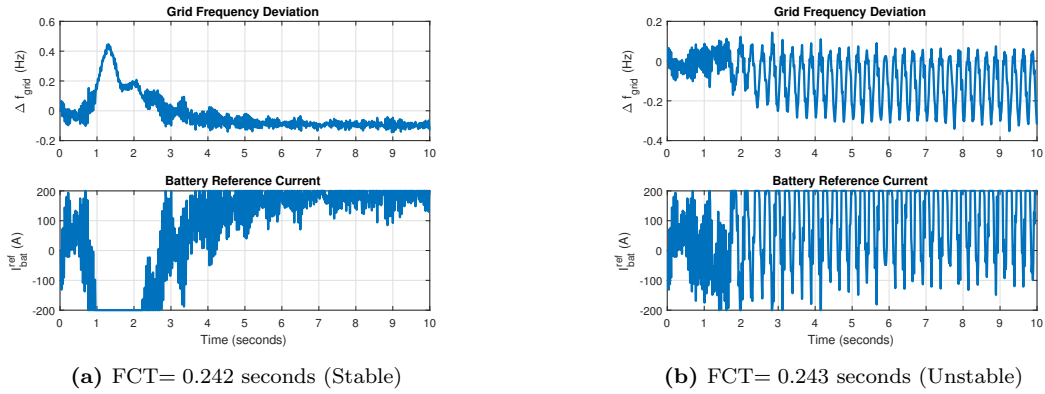


Figure A1. Frequency deviation and battery reference current of the local frequency control at active bus 2.

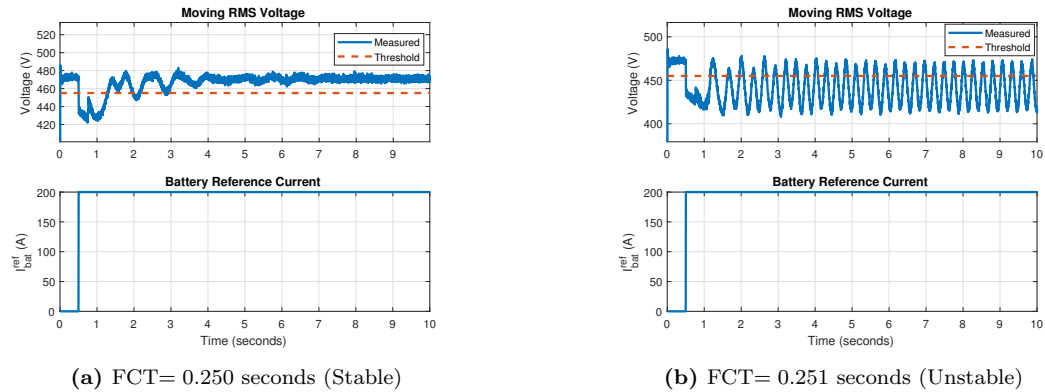


Figure A2. Moving RMS voltage fault detection and battery reference current of the constant battery current V2G at active bus 2.